

Why Python is Ideal for Machine Learning



Many programming languages facilitate machine learning, a few prominent ones being Java, Python, R and C++. Python leads the field in its ease of use and the simplicity of programming code. Here are some very compelling reasons for why one should use Python for machine learning.

Parents teach us behavioural skills right from childhood, teachers help us increase our knowledge of a range of subjects, and as adults we learn a few life skills based on our experiences. So learning is a part of our life, right from birth to death. Wouldn't it be amazing if our machines also start learning the same way we do? This question made Frank Rosenblatt jump into the arena of machine learning and neural networks in the 1950s, and he went on to develop the first neural network for computers, which is best known as the Perceptron.

We always wonder how machines can be made to learn by themselves—since they are non-living objects and do not have brains like us. So Rosenblatt tried giving brains to computers. He used the Perceptron to simulate different thought processes of the human brain, in computers. Our brains have a network of neurons and a central nervous system which helps us read situations, actions and conditions, and make decisions based on these. Similarly, the networks developed by Rosenblatt act like the network of neurons for machines or computers. The various algorithms used by neural networks act as the central nervous system, which helps machines make decisions on the basis of information or data collected.

In 1957, one of the first algorithms named Nearest Neighbour was written and it allowed computers to use basic pattern recognition techniques. This algorithm could be used

to map the route for travelling salesmen, starting at any city and ensuring they visited all the different cities they needed to, during their tour.

Today, machines are capable of learning by themselves, and this is called machine learning.

Machine learning (ML) is defined as a sub-set of artificial intelligence where different computer algorithms are used to autonomously learn from available data sets and information. In ML, machines or computers don't need to be explicitly programmed — they can change and continuously improve their algorithms by themselves. Nowadays, ML algorithms help computers to communicate with human beings on their own, write and publish sports match reports, autonomously drive cars, find terrorist suspects and do a whole lot more. ML gave birth to Sophia, a humanoid robot, which received citizenship from Saudi Arabia in 2017.

ML has basically evolved from the study of computational learning theory and pattern recognition in artificial intelligence. It explores the study and construction of different algorithms that can learn from and make predictions on the basis of the available data sets. All these algorithms follow the strictly static program instructions by applying data-driven decisions or predictions through the development of a model from given sample inputs. ML is implemented in a wide range of computing tasks where designing and developing explicit algorithms, that too, with good performance, is quite difficult and often not even feasible. For instance, e-mail filtering, the detection of network intruders or any malicious insiders working towards data breaches, learning to rank, optical character recognition, etc, are some well known applications of ML.

Types of machine learning

Machine learning is broadly classified into three categories on the basis of the way machines learn to increase their knowledge and take decisions.

Supervised machine learning: In this case, we train and teach the machine or computer, using available data sets that are

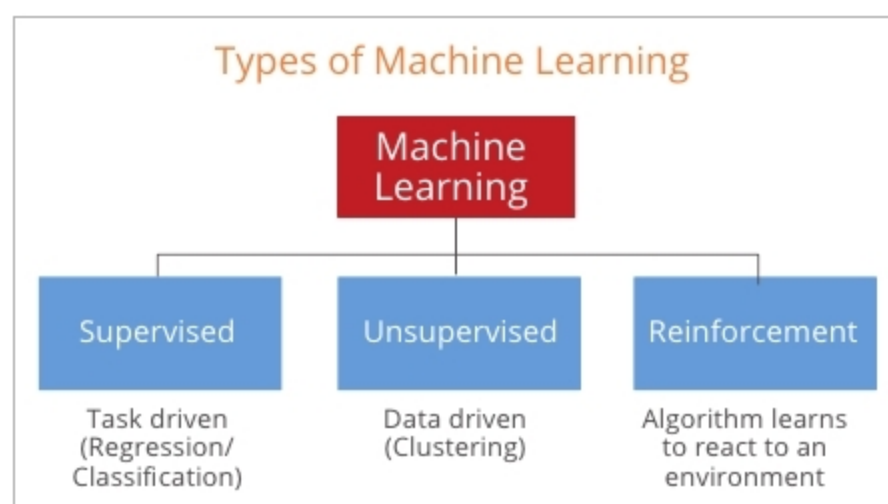


Figure 1: Different types of machine learning (Image source: [googleimages.com](https://www.googleimages.com))